Home Address: **42 Glen Road** **Sarisbury Green** **Southampton** **Hampshire** **SO31 7FF**	# Harry New **Mobile: 07767 414714** **Email: hn457@bath.ac.uk** **Nationality: British** **British Driving License**	**Term Address:** **27 Manor Road** **Bath** **Somerset** **BA1 4BW**

Education

University of Bath Department of Engineering **2021-2026**
MEng (Hons) Integrated Design Engineering (5 Year Sandwich)
Projected Mark: First-Class Honours

Peter Symonds College, Winchester – A Level Results **2019-2021**

Mathematics	**A***	Physics	**A**
Further Mathematics	**A**	Product Design	**A***

St Mary's Independent School – GCSE Results **2014-2019**
GCSEs: 4x Grade 9 including Mathematics and Physics, 4x Grade 8, 2x Grade 7, 1x Grade 6

Work Experience

Mercor – Contracted Software Developer
June 2025 to October 2025

Reviewed Python code generated by AI models for API and web application design across FastAPI and Django codebases. Assessed their adherence to RESTful design practices, including their focus on exposing resources, returning valid status codes, and using query parameters to filter the data effectively. Provided detailed technical feedback and suggested improvements to refine model outputs and created targeted prompts to generate further responses for evaluation.

Envisics – Junior Software Developer
June 2023 to September 2024

Created numerous internal tools and test scripts interfacing with the embedded system and end-of-line (EOL) testing systems. Made extensive use of Python to extract data regarding the onboard properties and process these externally, including applying filters (e.g. Gaussian Filter) and visualising the data using matplotlib. Used Python to vary onboard registers and testing equipment (e.g. cameras, thermocouples, climatic chambers) to create full test sequences, outputting the processed data in log files and spreadsheets. Developed C code to resolve a connection dropout issue during week-long tests on a Raspberry Pi-based continuous monitoring system running on Linux, authored unit tests for the existing codebase, and produced clear documentation to support future development. Retrieved EOL testing data from a REST API using Postman, handling both JSON and XML responses, before automating data retrieval to a local SQL database to enable efficient processing and visualisation. Used Git for version control and collaborative development.

Technical Projects and Skills

Rover Software and Simulation
February 2025 to May 2025

Designed a rover to compete in the Olympus Rover Trials (ORT) student competition, serving as Principal Software Engineer and System Design Lead, , following the V-model development process from requirements capture through to validation. Developed the full system architecture using SysML diagrams, defining interfaces between subsystems, including a Flask UI, a Simulink onboard control system, and a CoppeliaSim physical simulation model. Created a sequence diagram to establish timings for the bootup, operation, and communication across the entire rover. Implemented and tuned Ackermann steering and point-turn control algorithms in Simulink using the Ziegler-Nichols PID method for an independently steered 6-wheel rover. Delivered a fully

	functional digital Mars yard in CoppeliaSim, integrating the control system and CAD model to enable hardware-free validation against the real Airbus competition environment.
Rolling Robot and Docking Station Project October 2025 to November 2026	Led a team of five as Team Lead and Principal Systems Engineer to design and demonstrate a rolling ball robot capable of autonomously navigating to a docking station and achieving wireless charging. Coordinated development across both the robot and docking station, planning subsystem, system, and integrated tests for the full end-to-end system. Developed an omnidirectional control system in Simulink with PID controllers to deliver consistent torque across three omnidirectional wheels. Built a dual TCP communication architecture connecting the robot and docking station to a central mission control computer. Designed a computer vision pipeline to detect ArUco markers for navigation, transitioning to ultrasonic sensors for precise autonomous docking as the robot closed in. Successfully demonstrated full autonomous docking and wireless charging.
Magnet Detection and Marker Project November 2024 to December 2025	Developed an autonomous embedded system using an Arduino Mega 2560 and hall-effect sensors to detect and locate magnets within a region of interest. Implemented a data processing pipeline in MATLAB, including a convolution filter for signal smoothing, binary thresholding for region identification, and a distance transform with watershed algorithm to pinpoint magnet centres. Designed a marker placement algorithm with logic to avoid previously placed markers. Built a physical marker-placement mechanism driven by a stepper motor, using rotary encoders for positional feedback and limit switches for end-stop detection.

Computing Skills

Python	Proficient in Python with 3 years of experience, including 1 year in industry. Experienced with web frameworks (Django, FastAPI, Flask) and data science libraries (pandas, matplotlib, scikit-learn). Certified Entry-Level Python Programmer (PCEP-30-02).
MATLAB and Simulink	Proficient in MATLAB with 5 years of experience, including 1 year in industry. Experienced with the Signal Processing and Control Systems Toolboxes for data analysis, visualisation, and design optimisation. Proficient in Simulink with 1 year of experience, used for control system design targeting embedded systems.
Git	Proficient in Git with 3 years of experience across personal, university, and industry projects, including 1 year in industry. Experienced in collaborative workflows with teams of up to 3 developers, utilising Git Flow for branch management across features and fixes. Led repository setup and enforced code quality standards as Principal Software Engineer within university projects, conducting pull requests, code reviews, and resolving merge conflicts. Primarily used via GitHub.
SQL	Experienced in MySQL, with practical application across personal and industry projects. Developed Python-MySQL integrations for data querying and analysis, and independently set up a local MySQL database at Envisics to streamline access to EOL data.
Other Skills	Demonstrated ability to rapidly adopt new technologies as required, with project-based experience in HTML, CSS, JavaScript, C, C++, Java, and Godot.

Interests

Weightlifting and Calisthenics	I've been interested in weightlifting for many years, which is both enjoyable and physically challenging. More recently, I've started callisthenics, which has given me new motivation in the gym to achieve more demanding exercises.
Drums	I've played the drums since 7 years old and find them a good way to relax. I am currently Grade 4 and am working towards the higher grades.